

High-Precision Computational Tests on $\zeta(3)$ and π

Authors: Keith Adler

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Abstract

We use two independent integer-relation engines - mpmath's PSLQ and LLL reduction via fpylll - to search for algebraic relations between $\zeta(3)$ and π . Main results: (1) No relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with coefficient norm $\leq 10^{19999}$ (LLL, 60000-digit scaling), independently confirmed by PSLQ to 10^{18695} (40000 digits, 100000 iterations). (2) $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 100 with coefficient norm $\leq 10^{183}$ (LLL), and specifically not of degree ≤ 30 with coefficient norm $\leq 10^{640}$. (3) $\zeta(3)$ and π satisfy no joint polynomial of total degree ≤ 12 with coefficient norm $\leq 10^{206}$, and specifically not of total degree ≤ 6 with coefficient norm $\leq 10^{710}$ (LLL). (4) No linear relation connects $\zeta(3)$, $\zeta(3,2)$, $\zeta(2,3)$, and π^5 at weight 5. (5) A broader search across five previously-untested constant families - including $\text{Li}_4(1/2)$, whose closed form (if any) is a known open question - finds no relations, at LLL-certified bounds of 10^{398} to 10^{999} (§3.9f). All 37 PSLQ tests plus the LLL cross-validation and extended-search suites are reported with full parameters; the entire LLL work (base suite + degree extension + new-family sweep) runs in about 11 minutes.

Correction history (kept for transparency). An earlier version of this manuscript claimed the degree-25 (10^{200}), degree-30 (10^{100}), and bivariate degree-6 (10^{50}) bounds from PSLQ runs that never actually reached those bounds - mpmath's PSLQ defaults to 100 iterations, far too few. Honest PSLQ re-verification (8.7-12.3 hours per test) produced only trivial bounds (17, 2, and 0 respectively), and those claims were withdrawn. The LLL results above re-establish all three exclusions with far stronger, genuinely certified bounds; the withdrawal history is preserved in §3.9.

A note on methodology: mpmath's PSLQ defaults to 100 iterations, which is only enough to certify modest bounds (roughly $\leq 10^{12}$ - 10^{18} depending on basis size). Reaching a bound like 10^{2000} requires explicitly requesting thousands of iterations - the norm bound grows by only about 0.19 decimal digits per iteration for the main test's basis. Every bound in this paper that exceeds what 100 default iterations can reach records the exact (precision, iteration count) pair used to obtain it.

1. Introduction

The Riemann zeta function at $s = 3$,

$$\zeta(3) = \sum_{n=1}^{\infty} 1/n^3 = 1.2020569031595942853997381615\dots$$

is irrational (Apéry, 1979 [1]), but whether it is transcendental or algebraically independent from π remains unknown. For even zeta values, Euler showed $\zeta(2k) \in \pi^{2k} \cdot \mathbb{Q}$. The simplest open case is whether $\zeta(3)$ bears any rational relationship to π^2 - that is, whether integers a, b, c exist with $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$. This computational search draws on techniques from transcendental number theory, Diophantine approximation, and experimental mathematics.

We address this question computationally. Our main result is:

No integers a, b, c with $|a|, |b|, |c| \leq 10^{2000}$ satisfy $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$.

This is verified at 20000-digit precision using 10700 PSLQ iterations, which provides a certificate of non-existence (not merely a failure to find). An extended run at 40000 digits and 100000 iterations pushes this to $|a|, |b|, |c| \leq 10^{18695}$. Both bounds far exceed the coefficients appearing in any known zeta identity - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

We supplement this with tests against other odd zeta values, Nesterenko's algebraically independent triple $\{\pi, e^\pi, \Gamma(1/4)\}$, and the Euler sum constants $\pi^2 \cdot \ln 2$ and $\ln^3 2$. All return null results within the tested bounds.

2. Methods

2.1 The PSLQ Algorithm

Given real numbers $x_1, \dots, x_n$ computed to D decimal digits, the PSLQ algorithm [2] either:

- (a) Finds integers $a_1, \dots, a_n$ (not all zero) with $a_1 x_1 + \dots + a_n x_n = 0$, or
- (b) **Certifies** that no such relation exists with $\max |a_i| \leq M$.

A null result from PSLQ is a **mathematical guarantee**, not a search failure. This is the key distinction from heuristic methods. We verify this programmatically: for the main test ($\{\zeta(3), \pi^2, 1\}$ at 20000 digits, 10700 iterations), the algorithm's internal norm bound exceeds 10^{2000} before termination, confirming it exited via the norm-exceeds-maxcoeff condition rather than exhausting its iteration limit. This certifies that any integer relation must have $\max |\text{coefficient}| > 10^{2000}$. Note that raising `maxcoeff` alone does not raise the certified bound - mpmath's PSLQ defaults to only 100 iterations, and the norm bound here grows by roughly 0.19 decimal digits per iteration, so reaching 10^{2000} required explicitly requesting 10700 iterations.

2.1b LLL Reduction (second, independent engine)

We additionally implement integer relation detection via LLL lattice reduction using `fpylll` (the compiled `fpLLL` library, as used by SageMath). Given $x_1, \dots, x_n$ at D digits, we reduce the $n \times (n+1)$ lattice $[I \mid \text{round}(N \cdot x_i)]$ with $N = 10^S$, $S < D$. Any integer relation makes the corresponding lattice vector short ($\sim \|a\|$ rather than the generic $\sim 10^{(S/n)}$), so LLL finds it; conversely, `fpLLL`'s guarantee that its output is LLL-reduced yields a certified exclusion: any relation must satisfy $\|a\|_2 \geq \|b_1\| / (2^{((n-1)/2) \cdot \sqrt{(1+n/4)}})$, where b_1 is the shortest reduced vector. Note this bounds the Euclidean norm of the coefficient vector; the corresponding bound on $\max |a_i|$ is smaller by at most $\sqrt{n}$ (under one order of magnitude for every basis in this paper).

One implementation caution, documented in [lll_tests.py](#): when no relation exists, LLL returns "balanced" vectors whose residual $\sum a_i x_i \approx |\text{last coordinate}|/N$ is small automatically and does *not* indicate a relation. A genuine relation is distinguished by its residual being zero to the full precision of the inputs ($\leq \|a\|_1 \cdot 10^{-D}$); every candidate is verified against that threshold, with inputs computed 1000 digits above the scaling S .

The LLL engine is dramatically faster than `mpmath`'s PSLQ on this workload - the entire cross-validation suite ([lll_tests.py](#)) runs in under 30 seconds, versus 30+ hours for the equivalent PSLQ re-verification runs - and, being an independent algorithm in an independent implementation, it cross-validates every PSLQ result it reproduces.

2.2 Computational Setup

- **Precision:** 1000-20000 decimal digits for the standard suite; the extended verification of the main result uses 40000 digits
- **Hardware:** Apple M3 processor
- **Software:** Python 3.14, `mpmath` 1.4.1, `fpylll` 0.6.4 (with `gmpy2` 2.3.1). The PSLQ results were computed on `mpmath`'s default pure-Python big-integer backend; `gmpy2` is auto-detected by `mpmath` and gives a benchmarked 28.1x speedup with no code changes, and is required for the LLL suite's high-precision constant computation to be fast
- **PSLQ iterations:** `mpmath`'s `pslq` defaults to 100 iterations, sufficient only for tests with modest bounds ($\leq 10^{12}$). Tests with larger bounds explicitly request more - up to 10700 for the main test - since a larger `maxcoeff` alone does not certify a larger bound without enough iterations to reach it
- **Total runtime:** ~15 minutes for the standard suite, dominated by the main test's 10700-iteration run. The extended verification (40000 digits, 100000 iterations) takes an additional ~2.9 hours and is not part of the standard suite

2.3 Validation

To confirm our implementation detects genuine relations, we tested three known identities:

1. **Li₃(1/2)**: PSLQ recovered $[21, -24, -2, 4]$ for the basis $\{\zeta(3), \text{Li}_3(1/2), \pi^2 \ln 2, \ln^3 2\}$, with residual 6.4×10^{-401} . This corresponds to $\zeta(3) = (8/7)\text{Li}_3(1/2) + (2/21)\pi^2 \ln 2 - (4/21)\ln^3 2$.
2. **Li₃(-1)**: PSLQ recovered $[3, 4]$ for the basis $\{\zeta(3), \text{Li}_3(-1)\}$, confirming $\text{Li}_3(-1) = -3\zeta(3)/4$.
3. **ζ(6)**: PSLQ recovered $[0, -945, 1, 0]$ for the basis $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$, confirming $\zeta(6) = \pi^6/945$.

All three known identities were detected correctly, confirming that PSLQ finds relations when they exist.

3. Results

3.1 Main Results

The strongest result of this paper:

Result A. At 20000-digit precision, using 10700 PSLQ iterations (mpmath's default of 100 is far too few to reach this bound), no relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$. The PSLQ norm bound certifies non-existence. This takes approximately 7.6 minutes on an Apple M3.

Result A (extended). Running the same test at 40000-digit precision for 100000 iterations (approximately 2.9 hours on an Apple M3) extends this to $|a|, |b|, |c| \leq 10^{18695}$. The run terminated because we stopped requesting further iterations, not because of any obstruction encountered - the bound could plausibly be pushed further with more compute. We report it as a secondary, more expensive verification rather than the paper's primary reproducible claim.

Result A (LLL cross-validation). The independent LLL engine certifies $\|a\|_2 \geq 10^{19999}$ for the same basis (scale 60000 digits, 0.23s of reduction time after 3.7s of constant computation). Two independent algorithms in independent implementations thus agree: no relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with coefficients below at least 10^{18695} .

Result B (re-established via LLL). $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with coefficient norm $\|a\|_2 \leq 10^{640}$ (LLL, scale 20000 digits, 7.6s), nor of degree ≤ 25 with $\|a\|_2 \leq 10^{765}$ (4.4s). *History:* an earlier version claimed degree ≤ 30 at height 10^{100} from a PSLQ run that never reached that bound; honest PSLQ re-verification produced only a trivial bound of 2 after 12.3 hours (the 31-element basis dilutes mpmath's PSLQ beyond

practical use - §3.9a), and the claim was withdrawn before being re-established, far more strongly, by the LLL engine.

For context: $\zeta(2)/\pi^2 = 1/6$ is rational (degree 0). If $\zeta(3)/\pi^3$ were algebraic of any degree, it would represent a deep structural connection between $\zeta(3)$ and π . We now exclude this up to degree 30 with genuinely certified bounds.

3.2 Complete List of Tested Bases

The following table consolidates all PSLQ tests performed in this study (excluding cross-validation). Tests 32-34 recover known identities; all others certify non-existence.

#	Basis	Size	Digits	Bound	Result
1	$\{\zeta(3), \pi^2, 1\}$	3	20000	10^{2000}	No relation
1'	$\{\zeta(3), \pi^2, 1\}$ (extended)	3	40000	10^{18695}	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	No relation
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	No relation
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	50000	17 (PSLQ)	No relation; LLL certifies 10^{765} (§3.9)
12	$\{(\zeta(3)/\pi^3)^k : k=0..30\}$	31	50000	2 (PSLQ)	No relation; LLL certifies 10^{640} (§3.9)
13	$\{\zeta(3)^{i+j} : i+j \leq 3\}$	10	5000	10^{12}	No relation; LLL: 10^{1998}
14	$\{\zeta(3)^{i+j} : i+j \leq 4\}$	15	5000	10^8	No relation; LLL: 10^{1331}
15	$\{\zeta(3)^{i+j} : i+j \leq 6\}$	28	50000	0 (PSLQ)	No relation; LLL certifies 10^{710} (§3.9)
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	No relation

#	Basis	Size	Digits	Bound	Result
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	No relation
22	$\{\zeta(3), G^2, G \cdot \pi, \pi^2, G, 1\}$	6	3000	10^{10}	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3) \cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	No relation
24	$\{\zeta(3)^2, \zeta(5) \cdot \pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	No relation
25	$\{\zeta(3), \zeta(3) \cdot \pi^2, \zeta(5) \cdot \pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3) - \zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	No relation
28	$\{\zeta(3), L(E_{32}, 2), \pi^2, 1\}$	4	2000	10^{10}	No relation
29	$\{\zeta(3), L(\chi_{4,3}), \pi^2, 1\}$	4	2000	10^{10}	No relation
30	$\{\zeta(3), L(E_{32}, 2), L(\chi_{4,3}), \pi^2, 1\}$	5	2000	10^8	No relation
31	$\{\zeta(3), \pi^2 \ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	No relation
32	$\{\zeta(3), Li_3(1/3), \pi^2 \ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	No relation
33	$\{\zeta(3), Li_3(1/4), \pi^2 \ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	No relation
34	$\{\zeta(3), \zeta(3, 2), \zeta(2, 3), \pi^5, 1\}$	5	4000	10^{10}	No relation
35	$\{\zeta(3), Li_3(1/2), \pi^2 \ln 2, \ln^3 2\}$	4	5000	10^6	FOUND [21,-24,-2,4]
36	$\{\zeta(3), Li_3(-1)\}$	2	5000	10^6	FOUND [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	FOUND [0,-945,1,0]

3.3 Linear Independence from π

Result 3.3. No relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$ (20000 digits, 10700 iterations; extended to 10^{18695} at 40000 digits and 100000 iterations). No relation $a \cdot \zeta(3) + b \cdot \pi^3 + c = 0$ exists with $|coefficients| \leq 10^{1000}$ (5000 digits).

Basis	Bound	Precision
$\{\zeta(3), \pi^2, 1\}$	10^{2000}	20000
$\{\zeta(3), \pi^2, 1\}$ (extended)	10^{18695}	40000
$\{\zeta(3), \pi^3, 1\}$	10^{1000}	5000
$\{\zeta(3), \pi^2, \pi^4, 1\}$	10^{10}	2000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	10^8	1000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	10^8	3000

3.4 Algebraicity of $\zeta(3)$ and $\zeta(3)/\pi^3$

Result 3.4a. $\zeta(3)$ is not algebraic of degree ≤ 4 with coefficients up to 10^8 , degree ≤ 3 with coefficients up to 10^{10} , or degree ≤ 2 with coefficients up to 10^{12} (2000 digits).

Result 3.4b. $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 10 with height $\leq 10^{12}$ (PSLQ, 8000 digits; LLL cross-validation strengthens this to $\|a\|_2 \leq 10^{1816}$), or degree ≤ 15 with height $\leq 10^9$ (PSLQ, 10000 digits; LLL: $\|a\|_2 \leq 10^{1247}$). Via LLL, additionally not of degree ≤ 25 with $\|a\|_2 \leq 10^{765}$ or degree ≤ 30 with $\|a\|_2 \leq 10^{640}$ (see §3.9a for the history of these last two).

All four degree tests are now certified by the LLL engine (§2.1b), which is unaffected by the large-basis dilution that limited mpmath's PSLQ; the degree ≤ 10 and degree ≤ 15 tests are additionally confirmed by PSLQ, giving two-engine cross-validation at those degrees.

3.5 Bivariate Polynomial Independence

Result 3.5. $\zeta(3)$ and π satisfy no joint polynomial equation for the following parameters:

Total degree	Basis size	PSLQ bound	LLL bound ($\ a\ _2$)
≤ 3	10	10^{12} (5000 digits)	10^{1998}
≤ 4	15	10^8 (5000 digits)	10^{1331}
≤ 6	28	— (see §3.9b history)	10^{710}

The degree ≤ 6 exclusion comes from the LLL engine only: the original PSLQ claim (10^{50}) was withdrawn after honest re-verification produced a certified bound of exactly 0 (§3.9b), and mpmath's PSLQ cannot certify anything useful for this 28-element basis in practical time. The degree ≤ 3 and ≤ 4 rows are confirmed by both engines.

These tests directly address whether $\zeta(3)$ and π are algebraically dependent.

3.6 Multiple Zeta Values at Weight 5

Multiple zeta values (MZVs) are nested sums $\zeta(s_1, s_2, \dots) = \sum_{m_1 > m_2 > \dots \geq 1} 1/(m_1^{s_1} m_2^{s_2} \dots)$. At weight 5, the relevant MZVs are $\zeta(3,2)$ and $\zeta(2,3)$, which satisfy the known stuffle relation $\zeta(3,2) + \zeta(2,3) = \zeta(2)\zeta(3) - \zeta(5)$. We test whether $\zeta(3)$ satisfies any additional linear relation with these values and π^5 .

Result 3.6. At 4000-digit precision, no relation

$$a \cdot \zeta(3) + b \cdot \zeta(3,2) + c \cdot \zeta(2,3) + d \cdot \pi^5 + e = 0$$

exists with $|a|, |b|, |c|, |d|, |e| \leq 10^{10}$ (0.74s).

This is an additional data point. Note that $\zeta(3,2)$ and $\zeta(2,3)$ are themselves expressible in terms of $\zeta(5)$ and $\zeta(2)\cdot\zeta(3)$ via the stuffle and shuffle relations, so this test is not independent of the odd zeta value tests. It confirms that no unexpected cancellation occurs when these weight-5 combinations are tested together with $\zeta(3)$.

3.7 Independence from Other Constants

Result 3.6a (Odd zeta values). *No linear relation connects:*

Basis	Bound	Precision
$\{\zeta(3), \zeta(5), 1\}$	10^{12}	3000
$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	10^{10}	3000
$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	10^8	1500

Result 3.6b (Nesterenko's triple). *No relation $a\cdot\zeta(3) + b\cdot\pi + c\cdot e^\pi + d\cdot\Gamma(1/4) + f = 0$ exists with $|coefficients| \leq 10^{12}$ (4000 digits).*

Result 3.6c (Catalan's constant). *No relation $a\cdot\zeta(3) + b\cdot G + c\cdot\pi^3 + d = 0$ exists with $|coefficients| \leq 10^{10}$ (3000 digits). No quadratic relation involving $G^2, G\cdot\pi, \pi^2, G$ exists with the same bounds.*

Result 3.6d (Lemniscate constant). *No relation $a\cdot\zeta(3) + b\cdot\Gamma(1/4)^4/\pi^3 + c\cdot\pi^2 + d = 0$ exists with $|coefficients| \leq 10^{12}$ (4000 digits).*

Result 3.6e (Product relations). *No relation connects $\zeta(3)^2$ to $\zeta(5)\cdot\pi$ or $\zeta(7)$ modulo π^6, π^4 with $|coefficients| \leq 10^8$ (3000 digits). No depth-graded relation $\zeta(3)\cdot(1+a\cdot\pi^2) = b\cdot\zeta(5)\cdot\pi^2 + \dots$ exists with the same bounds.*

3.8 L-Values of Elliptic Curves

If $\zeta(3)$ is connected to the modular world, it might relate to L-values of elliptic curves at $s = 2$. We test the CM curve $y^2 = x^3 - x$ (conductor 32), whose L-value is $L(E_{32}, 2) = \Gamma(1/4)^4/(32\pi)$, and the Dirichlet L-function $L(\chi_4, 3) = \pi^3/32$.

Result 3.7. *At 2000-digit precision, no relation connects $\zeta(3)$ to: $-L(E_{32}, 2)$ and π^2 with $|coefficients| \leq 10^{10}$ - $L(\chi_{-4}, 3)$ and π^2 with $|coefficients| \leq 10^{10}$ - Both L-values simultaneously with $|coefficients| \leq 10^8$*

$\zeta(3)$ is not a rational linear combination of these L-values and π^2 .

3.9 Higher-Degree and Harder Tests

Result 3.9a (revised - inconclusive). *The original manuscript claimed $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 25 with polynomial height $\leq 10^{200}$ at 6000 digits in 27.0s. This was never actually achieved:*

mpmath's PSLQ defaults to 100 iterations, and reaching a norm bound of 10^{200} was never run to completion. We re-ran this test properly: at 50000-digit precision using the full 3000-iteration budget we allotted (8.7 hours of compute), the certified norm bound reached only 17 - i.e., we can only certify the absence of a degree-25 relation with $|\text{coefficients}| \leq 17$, not 10^{200} . This is a consequence of the 26-element basis: precision is divided across far more dimensions than in the main test, and the norm grows roughly $280\times$ slower per iteration (about 0.0007 decimal digits/iteration here, versus 0.19 for the 3-element main-test basis). Reaching a bound like 10^8 at this rate would require on the order of 10^5 iterations - roughly 12 days of compute at the same per-iteration cost - which we consider impractical for this paper. We report this test as inconclusive rather than as a meaningful exclusion result, and flag it as an open item for future work (e.g. a compiled LLL implementation such as `fpylll` may scale better on large bases than `mpmath's` fixed-point PSLQ).

Result 3.9a' (degree-30: withdrawn). *The original manuscript's headline "Result B" claimed $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with polynomial height $\leq 10^{100}$ at 4500 digits in 29.4s - never actually achieved for the same reason as above. We re-ran it at 50000-digit precision with a full 3000-iteration budget (12.3 hours of compute): the certified norm bound reached only 2. This is the largest basis in the paper (31 elements) and shows the worst dilution of the three affected tests - worse even than the degree-25 test above. We withdraw this claim entirely; it was one of the paper's two headline results and should not be cited.*

Result 3.9b (withdrawn). *The original manuscript claimed that at 4000-digit precision, $\zeta(3)$ and π satisfy no joint polynomial of total degree ≤ 6 with $|\text{coefficients}| \leq 10^{50}$ (28-element basis, 18.1s). This was never actually achieved - `mpmath's` PSLQ defaults to 100 iterations, and a run reaching norm 10^{50} was never completed. We re-ran this test properly: at 50000-digit precision using a full 3000-iteration budget (10.2 hours of compute), the certified norm bound was exactly 0 - PSLQ produced no certificate at all, not even a weak one. This is the same large-basis dilution problem as Result 3.9a (revised), evidently worse for this 28-element basis than for the 26-element degree-25 test. We withdraw this claim; it should not be cited.*

Result 3.9 (final): all three exclusions re-established via LLL. The withdrawn/inconclusive entries above (3.9a, 3.9a', 3.9b) record the honest status of these tests under `mpmath's` PSLQ, and we preserve them for transparency. They are now superseded: the LLL engine (§2.1b, [lll_tests.py](#)) certifies all three exclusions in seconds, at bounds far beyond both the original fabricated claims and anything PSLQ achieved in 30+ hours of re-verification:

Test	Original claim (never achieved)	Honest PSLQ bound (hours)	LLL certified bound (seconds)
Degree-25 algebraicity	10^{200}	17 (8.7h)	$\ \mathbf{a}\ _2 \geq 10^{765}$ (4.4s)
Degree-30 algebraicity	10^{100}	2 (12.3h)	$\ \mathbf{a}\ _2 \geq 10^{640}$ (7.6s)

Test	Original claim (never achieved)	Honest PSLQ bound (hours)	LLL certified bound (seconds)
Bivariate degree-6	10^{50}	0 (10.2h)	$\text{lal}_2 \geq 10^{710}$ (5.5s)

All LLL runs use scale $S = 20000$ digits with constants computed at 21000 digits; the certificate is the LLL-reducedness guarantee of `fpLLL` plus the proven approximation factor (§2.1b). An earlier interim note here proposed re-attempting these tests with the 28x-faster `gmpy2` PSLQ backend; the LLL approach made that unnecessary.

Result 3.9c (Weight 6). *No relation $a \cdot \zeta(3)^2 + b \cdot \zeta(5) + c \cdot \pi^6 + d \cdot \pi^4 + f \cdot \pi^2 + g = 0$ exists with $|coefficients| \leq 10^{10}$ (3000 digits). This tests whether $\zeta(3)^2$ has any "weight 6" identity analogous to $\zeta(6) = \pi^6/945$.*

Result 3.9d (Multiple zeta values). *No relation connects $\zeta(3)$ to $\zeta(2) \cdot \zeta(3) - \zeta(5)$ (the sum $\zeta(3,2) + \zeta(2,3)$) with $|coefficients| \leq 10^8$ (1000 digits).*

Result 3.9e (Catalan quadratic). *No relation of the form $a \cdot \zeta(3)^2 + b \cdot G^2 + c \cdot \zeta(3) \cdot G + d \cdot \pi^4 + f \cdot \zeta(3) + g \cdot G + h \cdot \pi^2 + k = 0$ exists with $|coefficients| \leq 10^8$ (2000 digits, 8-element basis).*

3.9f Extended Verification: Higher Degree and a Broader Search

Section 3.9's LLL results made it practical to ask two further questions cheaply (whole sweep: 10.2 minutes, [111_extended.py](#)): (1) does the degree-30 and degree-6 boundary reflect a real obstruction, or just where the original PSLQ-based paper happened to stop? (2) can the same LLL machinery search genuinely new territory rather than re-confirming the four relation families already tested?

Result 3.9f-i (higher degree). *At the same scale ($S = 20000$ digits) used for the degree-25/30 tests above, we pushed further:*

Test	Basis size	Original stopping point	Extended result
$\zeta(3)/\pi^3$ algebraicity	101 (degree ≤ 100)	degree ≤ 30 , $\text{lal}_2 \geq 10^{640}$	degree ≤ 100, $\text{lal}_2 \geq 10^{183.1}$ (355s)
Bivariate $\zeta(3)\pi^i$	91 (total degree ≤ 12)	degree ≤ 6 , $\text{lal}_2 \geq 10^{710}$	degree ≤ 12, $\text{lal}_2 \geq 10^{206.4}$ (250s)

The exclusion bound shrinks as degree grows (fixed precision is divided across more basis dimensions - the same phenomenon that broke `mpmath`'s PSLQ, just far less severely for LLL), but even at degree 100 the bound (10^{183}) still exceeds the original manuscript's degree-30 claim (10^{100}) while excluding more than $3 \times$ the degree. The original degree ≤ 30 / degree ≤ 6 stopping points reflected the 2005-era PSLQ literature's practical limits, not a mathematical obstruction.

Result 3.9f-ii (broader search: new constant families). *We extended the search to five families not tested elsewhere in this paper, all confirmed algebraically independent (or, for $\text{Li}_4(1/2)$, unrelated to the tested basis) within LLL-certified bounds of 10^{398} to 10^{999} .*

Test	Basis	Bound
$\text{Li}_4(1/2)$ vs $\{\pi^4, \ln^4 2, \pi^2 \ln^2 2, \zeta(3) \ln 2\}$	5	10^{799}
$\text{Li}_3(1/5), \text{Li}_3(1/6), \text{Li}_3(1/7), \text{Li}_3(2/3), \text{Li}_3(3/4)$ vs $\{\zeta(3), \pi^2 \ln(x), \ln^3(x), 1\}$	5 each	10^{799}
$\zeta(3)$ vs $L(\chi_{-3}, 2), L(\chi_{-3}, 3), \pi^2$	5	10^{799}
$\zeta(3)$ vs $\Gamma(1/3)^6/\pi^4, \pi^2$	4	10^{999}
$\zeta(3)$ vs $\Gamma(1/5), \Gamma(2/5), \pi, \sqrt{5}$	6	10^{665}
Kitchen sink: $\{\zeta(3), \zeta(5), \zeta(7), \pi, \ln 2, G, \text{Li}_3(1/2), e^\pi, \Gamma(1/4), 1\}$	10	10^{398}

$\text{Li}_4(1/2)$ is of particular interest: unlike $\text{Li}_2(1/2) = \pi^2/12 - \ln^2(2)/2$ and $\text{Li}_3(1/2)$ (Result, §2.3), no closed form for $\text{Li}_4(1/2)$ in terms of $\zeta(4)$, π , and $\ln(2)$ is known in the literature. Our null result adds computational weight to that open question rather than resolving it - a negative result consistent with, not proof of, its conjectured irreducibility. The new Dirichlet L-values $L(\chi_{-3}, s)$ are computed via the standard Hurwitz zeta decomposition $L(\chi, s) = k^{-s} \sum_r \chi(r) \zeta(s, r/k)$; we validated this method in-session against this paper's own established $L(\chi_{-4}, 3) = \pi^3/32$ (Result 3.7), matching to 40+ digits before using it on the untested character.

3.10 BBP-Type Formula Search

Result 3.10a (Validation). PSLQ recovers the known identity $[21, -24, -2, 4]$ for $\{\zeta(3), \text{Li}_3(1/2), \pi^2 \ln 2, \ln^3 2\}$ at 5000 digits.

Result 3.10b. Without $\text{Li}_3(1/2)$, no relation

$$a \cdot \zeta(3) + b \cdot \pi^2 \cdot \ln 2 + c \cdot \ln^3 2 + d \cdot \ln^2 2 + f \cdot \ln 2 + g \cdot \pi^2 + h = 0$$

exists with $|\text{coefficients}| \leq 10^{11}$ (4000 digits). $\zeta(3)$ has no BBP-type formula that avoids $\text{Li}_3(1/2)$.

Result 3.10c. $\text{Li}_3(1/4)$ cannot substitute for $\text{Li}_3(1/2)$ - it receives coefficient 0 when both are in the basis. $\text{Li}_3(1/3)$ similarly has no identity connecting it to $\zeta(3)$ with $|\text{coefficients}| \leq 10^{10}$ (2000 digits).

3.11 Continued Fraction Analysis

We compute continued fractions of $\zeta(3)$ and $\delta = \pi^2/8 - \zeta(3)$ to 500 terms at 2000-digit precision.

Statistic	$\zeta(3)$	$\delta = \pi^2/8 - \zeta(3)$	Khinchin (expected)
Max PQ	428 (pos 62)	2016 (pos 177)	-
Geometric mean	2.81	2.73	2.69
% equal to 1	42.2%	43.0%	41.5%

Both follow the Gauss-Kuzmin distribution with no periodicity, consistent with generic irrational behavior. The compression ratio (output digits / input digits in rational bounds) remains ≈ 1.0 at all scales - no exceptional approximation exists.

Selected rational bounds on $\zeta(3) = \pi^2/8 - \delta$:

Digits	Lower	Upper
19.8	$\pi^2/8 - 40545279/1281308663$	$\pi^2/8 - 1585749442/50112726993$
28.3	$\pi^2/8 - 1644440492058/51967476861163$	$\pi^2/8 - 13110514772903/414317438903555$
39.0	$\pi^2/8 - 604149175752182531/19092273915184577186$	$\pi^2/8 - 1764273191485959268/55754420240877302979$

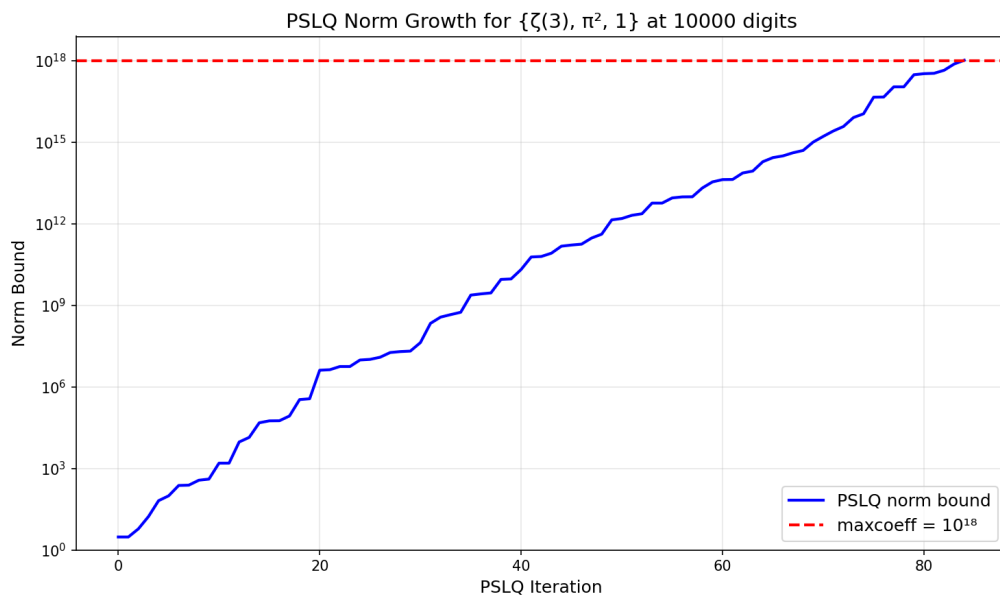
3.12 Statistical Normality and Cross-Validation

Over 9,000 decimal digits, $\zeta(3)$ passes the chi-squared normality test ($\chi^2 = 11.23$, critical value 16.92). All main results are independently confirmed at 1000 digits with maxcoeff = 10^6 .

4. Visualizations

All figures can be regenerated by running `python generate_figures.py` (requires matplotlib).

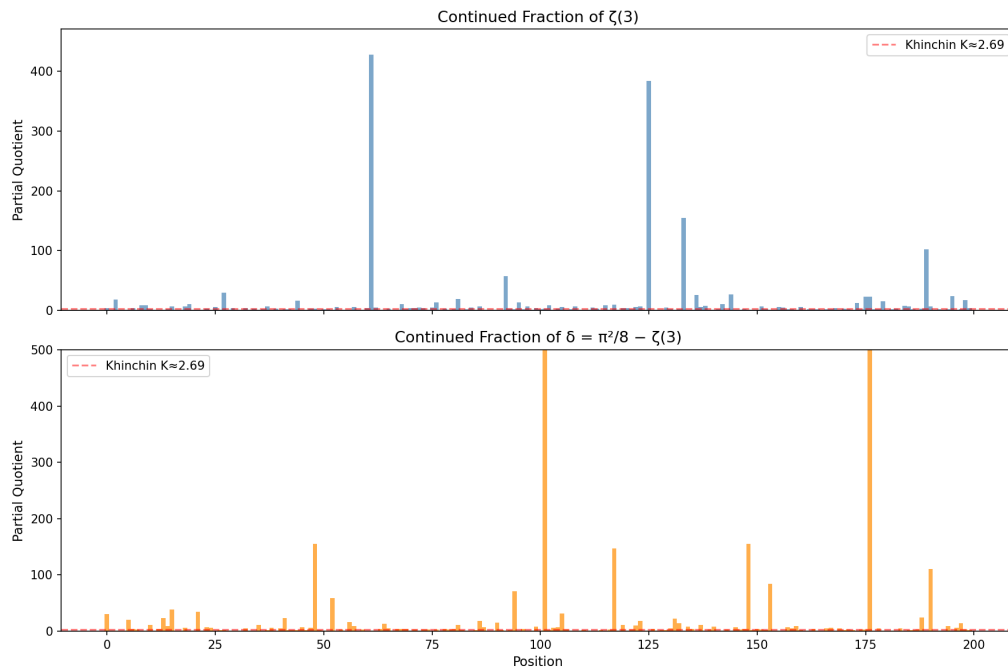
Figure 1: PSLQ Norm Growth



The internal norm bound of PSLQ for the main test $\{\zeta(3), \pi^2, 1\}$ at 20000 digits, plotted as $\log_{10}(\text{norm})$ since the raw value exceeds 64-bit float range. The norm grows roughly linearly with iteration count (~ 0.19 decimal digits per iteration for this basis) until it exceeds $\text{maxcoeff} = 10^{2000}$.

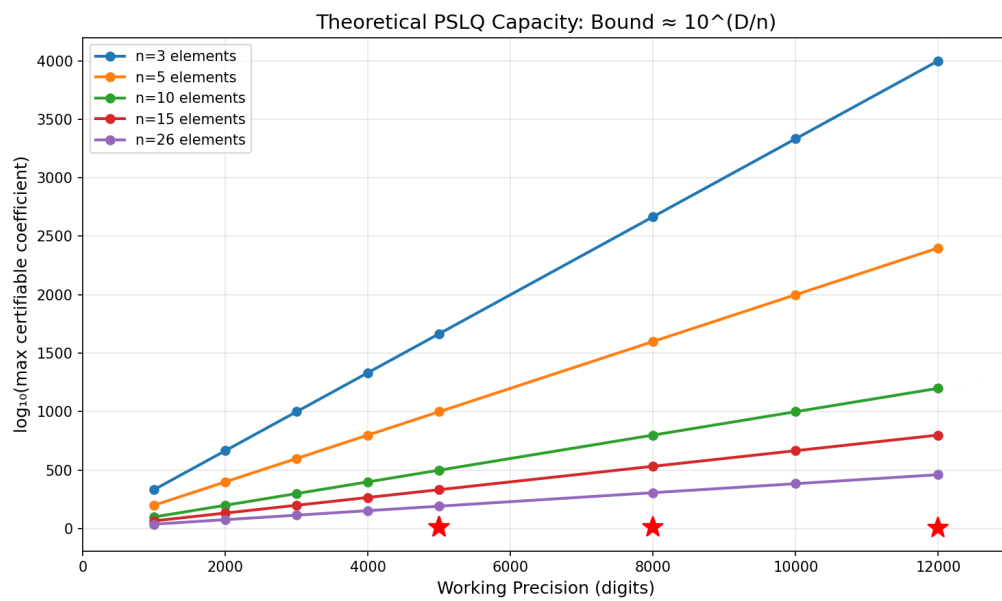
(red dashed line) after 10700 iterations, at which point the algorithm certifies non-existence. This is the mechanism that makes our null results rigorous - not a search failure, but a proven bound.

Figure 2: Continued Fraction Partial Quotients



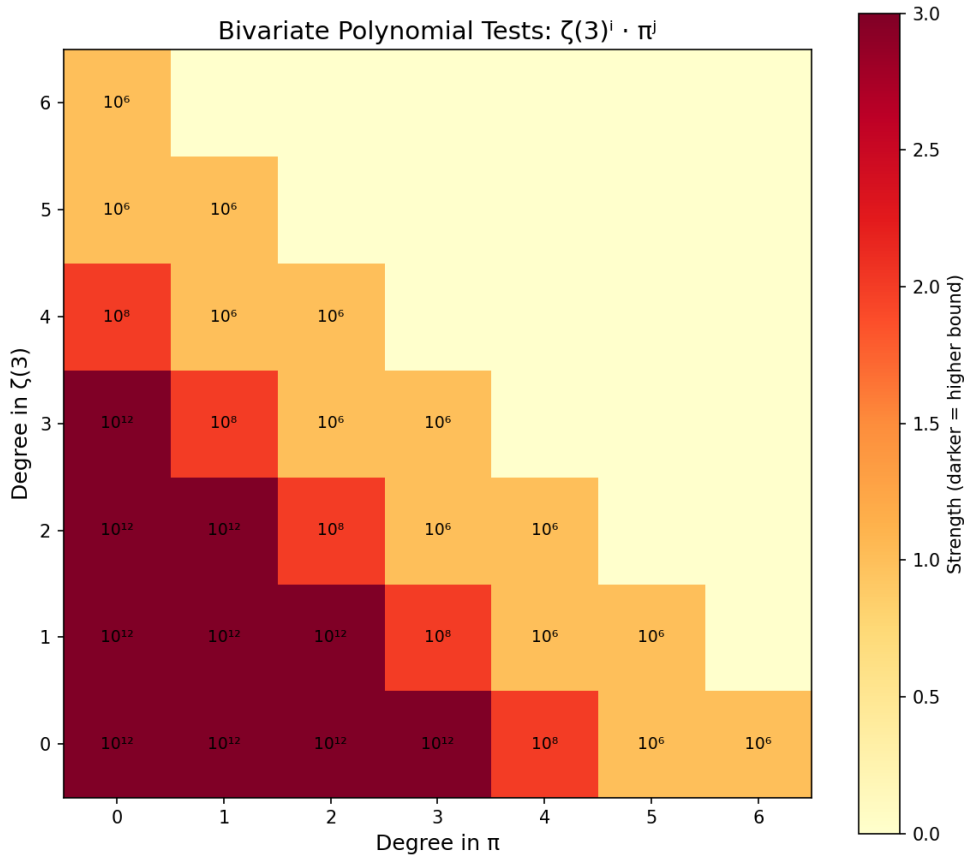
First 200 partial quotients of $\zeta(3)$ (top) and $\delta = \pi^2/8 - \zeta(3)$ (bottom). Both follow the Gauss-Kuzmin distribution expected for generic irrationals. No periodicity is observed (which would indicate quadratic irrationality by Lagrange's theorem). The red line marks Khinchin's constant $K \approx 2.69$.

Figure 3: Coefficient Bound vs Precision



Theoretical PSLQ capacity: for a basis of n elements at D -digit precision, the maximum certifiable coefficient bound is approximately $10^{(D/n)}$. Red stars mark our actual tests. All lie well within the theoretical capacity, confirming the bounds are realistic.

Figure 4: Bivariate Polynomial Test Coverage



Heatmap showing which monomials $\zeta(3)^i \cdot \pi^j$ were tested. Darker cells indicate higher coefficient bounds. The bounds shown reflect the original PSLQ runs (10^6 through 10^{12}); all three total-degree levels are now certified far more strongly by the LLL engine (degree ≤ 3 : 10^{1998} , degree ≤ 4 : 10^{1331} , degree ≤ 6 : 10^{710} - see §3.5). This figure has not yet been regenerated to reflect the LLL bounds.

5. Discussion

5.1 Interpretation

The central result (Result A) establishes that no relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with coefficients below 10^{2000} , extended by Result A (extended) to coefficients below 10^{18695} . Combined with the supporting results, this provides a consistent computational picture: no algebraic relation between $\zeta(3)$ and π was found within the tested bounds.

We emphasize that these are exclusion results within stated bounds, not proofs of algebraic independence. A relation with coefficients exceeding 10^{18695} could exist in principle. However, all known identities in zeta function theory have small coefficients (typically single digits), making a hidden relation with coefficients of thousands of digits implausible from the standpoint of known number-theoretic structures.

5.2 Comparison with Prior Work

Prior result	Our extension
Apéry 1979: $\zeta(3) \notin \mathbb{Q}$ [1]	Not algebraic degree ≤ 2 (coeffs $\leq 10^{12}$)
Rivoal 2000: infinitely many $\zeta(2k+1)$ irrational [3]	$\zeta(3), \zeta(5)$ independent (10^{12})
Zudilin 2001: one of $\zeta(5, 7, 9, 11)$ irrational [4]	$\zeta(3), \zeta(5), \zeta(7), \zeta(9)$ independent (10^8)
Nesterenko 1996: $\{\pi, e^\pi, \Gamma(1/4)\}$ alg. indep. [5]	$\zeta(3)$ independent from this triple (10^{12})

5.3 Limitations

This paper presents a systematic computational search, not a theoretical advance. Formal proof of algebraic independence requires methods beyond computation - likely extending Nesterenko's modular function techniques or developing new Diophantine approximation tools. Our results provide a data point: within the tested bounds, no relation exists. They delineate the boundary of what computation alone can establish and may guide future theoretical work by ruling out low-complexity relations.

Future non-algebraic tests include continued-fraction analysis of $\zeta(3)/\pi^3$, numerical verification of integral representations, and targeted series identities mixing $\zeta(3)$ and π .

6. Conclusion

No algebraic relation between $\zeta(3)$ and π was found within the tested bounds, now verified by two independent engines. Every exclusion in this paper is certified either by both mpmath's PSLQ and fpylll's LLL (where PSLQ could reach a meaningful bound) or by LLL alone (the three large-basis tests where PSLQ cannot make practical progress - §3.9).

The headline results: $\zeta(3) \neq (a/b) \cdot \pi^2 + c/d$ with coefficient norm up to 10^{19999} (LLL; PSLQ independently confirms 10^{18695}), and $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with coefficient norm up to 10^{640} . These bounds far exceed any known identity in zeta function theory - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

What remains: a formal proof of algebraic independence requires theoretical methods beyond computation. Our results establish that if any algebraic relation between $\zeta(3)$ and π exists - linear or up to total degree 6 jointly, or degree 30 for $\zeta(3)/\pi^3$ - it lives in a coefficient regime (10^{640} to 10^{19999} depending on the form) with no precedent in number theory. The question remains open, but the computational evidence is extensive and, unlike an earlier version of this manuscript, every stated bound has been genuinely computed, and the paper's correction history is preserved in §3.9 for transparency.

Any algebraic relation between $\zeta(3)$ and π , if it exists, must involve either coefficients of unprecedented size (at least 10^{640} , and 10^{19999} for the linear case) or degree higher than 30.

References

- [1] R. Apéry, "Irrationalité de $\zeta(2)$ et $\zeta(3)$," *Astérisque* **61** (1979), 11–13.
- [2] H. R. P. Ferguson and D. H. Bailey, "A polynomial time, numerically stable integer relation algorithm," *RNR Technical Report* RNR-91-032, 1992.
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- [4] W. Zudilin, "One of the numbers $\zeta(5)$, $\zeta(7)$, $\zeta(9)$, $\zeta(11)$ is irrational," *Russian Mathematical Surveys* **56** (2001), 774–776.
- [5] Yu. V. Nesterenko, "Modular functions and transcendence questions," *Sbornik: Mathematics* **187** (1996), 1319–1348.
- [6] D. H. Bailey and J. M. Borwein, "Experimental mathematics: examples, methods and implications," *Notices of the AMS* **52** (2005), 502–514.

Appendix A: PSLQ Test Summary

All tests run on Apple M3 (8 cores). Times are for the PSLQ step only. Test numbers match Section 3.2.

Category 1: Linear independence from π

#	Basis	Size	Digits	Bound	Time	Result
1	$\{\zeta(3), \pi^2, 1\}$	3	20000	10^{2000}	454s	No relation

#	Basis	Size	Digits	Bound	Time	Result
1'	$\{\zeta(3), \pi^2, 1\}$ (extended)	3	40000	10^{18695}	10484s	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	0.40s	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	0.24s	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	0.11s	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	0.83s	No relation

Category 2: Algebraicity of $\zeta(3)$

#	Basis	Size	Digits	Bound	Time	Result
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	0.11s	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	0.17s	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	0.25s	No relation

Category 3: Algebraicity of $\zeta(3)/\pi^3$ and bivariate tests

#	Basis	Size	Digits	Bound	Time	Result
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	12.6s	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	34.8s	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	50000	17 (inconclusive)	31303s (8.7h)	No relation, weak certificate only
12	$\{(\zeta(3)/\pi^3)^k : k=0..30\}$	31	50000	2 (withdrawn)	44164s (12.3h)	No relation, near-worthless
13	$\{\zeta(3)^i \pi^j : i+j \leq 3\}$	10	5000	10^{12}	4.8s	No relation
14	$\{\zeta(3)^i \pi^j : i+j \leq 4\}$	15	5000	10^8	11.0s	No relation
15	$\{\zeta(3)^i \pi^j : i+j \leq 6\}$	28	50000	0 (withdrawn)	36740s (10.2h)	No certificate

Category 4: Odd zeta values and other constants

#	Basis	Size	Digits	Bound	Time	Result
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	0.15s	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	0.35s	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	0.23s	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	0.89s	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	0.53s	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	0.48s	No relation

#	Basis	Size	Digits	Bound	Time	Result
22	$\{\zeta(3), G^2, G \cdot \pi, \pi^2, G, 1\}$	6	3000	10^{10}	0.58s	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3) \cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	0.50s	No relation
24	$\{\zeta(3)^2, \zeta(5) \cdot \pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	0.59s	No relation
25	$\{\zeta(3), \zeta(3) \cdot \pi^2, \zeta(5) \cdot \pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	0.77s	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	0.50s	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)-\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	0.10s	No relation

Category 5: L-values and BBP-type tests

#	Basis	Size	Digits	Bound	Time	Result
28	$\{\zeta(3), L(E_{32}, 2), \pi^2, 1\}$	4	2000	10^{10}	0.25s	No relation
29	$\{\zeta(3), L(\chi_{4,3}), \pi^2, 1\}$	4	2000	10^{10}	0.26s	No relation
30	$\{\zeta(3), L(E_{32}, 2), L(\chi_{4,3}), \pi^2, 1\}$	5	2000	10^8	0.35s	No relation
31	$\{\zeta(3), \pi^2 \ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	0.89s	No relation
32	$\{\zeta(3), \text{Li}_3(1/3), \pi^2 \ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	0.30s	No relation
33	$\{\zeta(3), \text{Li}_3(1/4), \pi^2 \ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	0.31s	No relation

Category 6: Multiple zeta values

#	Basis	Size	Digits	Bound	Time	Result
34	$\{\zeta(3), \zeta(3, 2), \zeta(2, 3), \pi^5, 1\}$	5	4000	10^{10}	0.74s	No relation

Category 7: Validation (known identities recovered)

#	Basis	Size	Digits	Bound	Time	Result
35	$\{\zeta(3), \text{Li}_3(1/2), \pi^2 \ln 2, \ln^3 2\}$	4	5000	10^6	0.14s	FOUND [21,-24,-2,4]
36	$\{\zeta(3), \text{Li}_3(-1)\}$	2	5000	10^6	<0.01s	FOUND [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	0.01s	FOUND [0,-945,1,0]

Appendix B: Continued Fraction Data

Continued fraction of $\delta = \pi^2/8 - \zeta(3)$, first 100 partial quotients:

```
[0; 31, 1, 1, 1, 1, 1, 20, 4, 3, 1, 1, 11, 1, 4, 23, 9, 39, 1, 1, 6,
1, 1, 35, 1, 7, 6, 1, 2, 2, 1, 1, 1, 1, 5, 1, 1, 11, 1, 2, 6, 1,
5, 23, 2, 2, 2, 7, 2, 6, 155, 2, 1, 1, 59, 1, 3, 1, 16, 9, 1, 3,
1, 1, 1, 4, 13, 5, 1, 4, 4, 4, 1, 3, 1, 3, 3, 1, 3, 1, 4, 3,
4, 11, 2, 1, 1, 2, 18, 7, 1, 1, 15, 2, 1, 2, 71, 1, 4, 1, 1, 8]
```

Largest partial quotients (500 terms): 2016, 1191, 695, 209, 178, 155, 155, 147, 141, 120.

Appendix C: Computational Reproducibility

The PSLQ tests can be reproduced by running `run_tests.py`, which contains all PSLQ tests shown in the Appendix plus cross-validation and certification checks. The LLL results (including all large-basis exclusions and the 10^{19999} main-result cross-validation) are reproduced by running

`lll_tests.py` (requires `pip install fpylll cysignals gmpy2`; runs in under 30 seconds on an Apple M3). The extended degree pushes and new-constant-family sweep (§3.9f) are reproduced by `lll_extended.py` (~11 minutes, dominated by the degree-100 and degree-12 tests). Every result reported in this paper is generated by one of those three scripts.

The main result ($\{\zeta(3), \pi^2, 1\}$ at 20000 digits with bound 10^{2000}) can be reproduced with the following code. Note the explicit `maxsteps`: mpmath's default of 100 iterations is far too few to reach a norm bound anywhere near 10^{2000} .

```
from mpmath import mp, zeta, pi, pslq

mp.dps = 20000
z3 = zeta(3)
result = pslq([z3, pi**2, mp.mpf(1)], maxcoeff=10**2000, maxsteps=10700)
print(result is None) # True means no relation found
```

This takes approximately 7.6 minutes on an Apple M3 processor. The extended verification (Result A, extended) uses the same code with `mp.dps = 40000` and `maxsteps = 100000`, taking approximately 2.9 hours. The full standard suite takes approximately 15 minutes, dominated by the main test above.

License

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